FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 — SUB-STRAND 2.1: PROPERTIES OF WAVES	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION ASSESSMENT — "Why does the matatu's horn change pitch as it passes, and why does the FM radio signal disappear inside the Kabete tunnel — and what do these two mysteries reveal about how ALL waves behave?"

WHAT IS THIS DOCUMENT?

This is your Final Explanation task. Over the past 12 lessons, you have investigated the anchoring phenomenon: the changing pitch of a matatu horn on the Nakuru–Nairobi highway and the disappearing FM radio signal inside the Kabete tunnel. Now it is time to bring all of your evidence, models, and reasoning together into one coherent scientific explanation.

HOW TO USE THIS DOCUMENT:

Read each section prompt carefully before you write. Use your Driving Question Board (DQB) notes, your lab data, your wave diagrams, and your revised models from throughout the unit. Each section builds on the one before it — your explanation should read as one connected argument, not a list of separate answers. Use scientific vocabulary correctly, connect every claim to evidence, and always link your reasoning back to the two real phenomena you observed at the start of this unit.

TIPS FOR SUCCESS:

- Make a claim → support it with specific evidence → explain the reasoning that connects them (C-E-R framework).
- Include labelled wave diagrams where helpful — a strong diagram can replace many sentences of description.
- Use Kenyan examples from this unit (the matatu, Classic 105 FM, the Kabete tunnel, the Rift Valley topography) to ground your explanation in real context.
- Write in full sentences and organise your ideas clearly within each section.
- Target length: 600–900 words total across all sections (not counting diagrams).

Section 1: The Universal Language of Waves — Core Properties	
Before you can explain either mystery, you need to establish the shared foundation. In this	All waves — whether mechanical like sound or electromagnetic like radio — transfer energy from one place to another without permanently moving the medium through which they travel. Every wave can be described using four key properties. Amplitude is the maximum displacement of the medium from its rest position; for a sound wave from the matatu horn, a larger amplitude means

section, define and explain the key properties that ALL waves share — whether they are sound waves from a matatu horn or radio waves from a Classic 105 FM transmitter in Nairobi. Your explanation must include: (a) amplitude, wavelength, frequency, and wave speed — with correct units and the relationship between them ($v = f\lambda$); (b) the distinction between transverse and longitudinal waves, with one example of each from the anchoring phenomenon; and (c) a clearly labelled wave diagram showing at least wavelength and amplitude. Explain in your own words why these properties matter — what does each one actually tell us about the wave?	a louder sound. Wavelength (λ) is the distance between two successive crests (or compressions), measured in metres. Frequency (f) is the number of complete wave cycles passing a fixed point every second, measured in Hertz (Hz). Wave speed (v) is how fast the wave pattern moves through the medium, measured in m/s. These three quantities are linked by the wave equation: $v = f\lambda$. This means that for a wave travelling at a fixed speed, a higher frequency must correspond to a shorter wavelength — a relationship that will be central to explaining the Kabete tunnel mystery. Sound waves, like the matatu horn, are longitudinal: the air particles vibrate parallel to the direction the wave travels, creating alternating compressions and rarefactions. Radio waves, like the Classic 105 FM signal, are transverse electromagnetic waves: the electric and magnetic fields oscillate perpendicular to the direction of travel, requiring no medium at all — which is why they can travel from a transmitter tower through the air to your radio receiver. [A labelled diagram here should show one full wavelength with amplitude marked on a transverse wave, and a separate diagram of a longitudinal wave showing compressions and rarefactions.]

Section 2: Mystery 1 — The Shifting Pitch of the Matatu Horn (The Doppler Effect)	
Now explain the first mystery directly. A matatu travelling in the opposite direction on the Nakuru–Nairobi highway sounds higher-pitched as it approaches and lower-pitched as it moves away — even though the driver never touches the horn. Use your understanding of wave properties to construct a full scientific explanation. Your explanation must: (a) identify the specific wave property that determines pitch; (b) explain, using a diagram or clear reasoning, why the relative motion between the matatu and the student changes that property; (c) use the correct	The pitch you hear from a sound source depends on the frequency of the sound waves reaching your ears — higher frequency means higher pitch. When the matatu is stationary, its horn emits sound waves at a constant frequency in all directions, and a listener anywhere nearby hears the same pitch. But when the matatu moves toward you at high speed along the Nakuru–Nairobi highway, something important happens: each successive compression of the sound wave is emitted from a position slightly closer to you than the one before. This squashes the wave crests together in front of the matatu, effectively shortening the wavelength on the approaching side. Since the speed of sound in air is approximately constant (about 340 m/s), a shorter wavelength means a higher frequency — and you hear a higher pitch. As the matatu passes and moves away, the opposite happens: each compression is emitted from a position slightly farther from you, stretching the wavelength out and lowering the frequency — so you hear a sudden drop in pitch. This phenomenon is called the Doppler Effect. It depends on the relative velocity between the source and the observer; the greater the speed of the matatu, the larger the shift in pitch. A second Kenyan example: athletes and spectators at the Nyayo National Stadium in Nairobi notice that the pitch of a referee's whistle sounds slightly higher as the referee sprints toward the far end of the pitch. Astronomers also use the Doppler Effect to measure how fast distant stars are moving relative to Earth — a technique that works because light is also a wave.

scientific name for this effect and state what it depends on; and (d) give one additional real-world example of the same effect that is relevant to Kenya (not the matatu itself).	

Section 3: Mystery 2 — The Kabete Tunnel and the Selective Signal Blackout	
Now explain the second mystery. When the student's bus enters the Kabete tunnel, the FM radio signal (Classic 105, 105.2 MHz) cuts out almost completely, but the AM signal remains audible. Both signals are electromagnetic waves broadcast from Nairobi — so why does the tunnel affect them differently? Your explanation must address: (a) the key difference between FM and AM signals in terms of wave properties (frequency and wavelength — use $v = f\lambda$ and the speed of light, 3×10^8 m/s, to calculate approximate wavelengths for each); (b) how the physical dimensions of the tunnel interact with the wavelength of each signal; (c) the role of wave behaviour — specifically reflection, absorption, and diffraction — inside the tunnel; and (d) why wavelength is the critical variable, not the type of broadcast technology.	Both FM and AM radio signals are electromagnetic waves travelling at the speed of light ($c = 3 \times 10^8$ m/s), but they differ significantly in frequency and therefore in wavelength. Classic 105 FM broadcasts at 105.2 MHz (105.2×10^6 Hz). Using $v = f\lambda$, its wavelength is $\lambda = (3 \times 10^8) \div (105.2 \times 10^6) \approx 2.85$ m. A typical AM station broadcasts at, say, 900 kHz (9×10^5 Hz), giving a wavelength of $\lambda = (3 \times 10^8) \div (9 \times 10^5) \approx 333$ m. The Kabete tunnel is a relatively confined structure — its walls, ceiling, and floor are made of reinforced concrete, which absorbs and reflects electromagnetic waves. The FM signal, with a wavelength of about 2.85 m, is comparable in size to the tunnel's cross-sectional dimensions. Waves behave very differently depending on how their wavelength compares to any obstacle or opening they encounter. FM waves are largely absorbed by the concrete walls and cannot diffract (bend) around corners or through the tunnel opening effectively — they require a relatively clear line-of-sight path to the transmitter. The result is a dramatic signal drop the moment the bus enters. The AM signal, with a wavelength of hundreds of metres — far larger than any feature of the tunnel — diffracts much more readily around obstacles and even follows the curvature of the ground. Its much longer wavelength means it is far less affected by the tunnel's physical dimensions, allowing it to remain audible even inside. This is why wavelength — not simply whether a signal is 'FM' or 'AM' — is the key variable: it determines how the wave interacts with its physical environment.

Section 4: The Unifying Principle — What Both Mysteries Reveal About ALL Waves	
You have now explained both mysteries separately. In this section, step back and draw the	Both mysteries, though they seem unrelated at first, are explained by the same set of wave properties — demonstrating that sound and electromagnetic waves, despite their differences, obey the same fundamental rules. First, both are described by frequency, wavelength, and wave speed, all connected by $v = f\lambda$: this single equation allowed us to calculate the FM wavelength that

big picture. What do the matatu horn and the Kabete tunnel blackout — taken together — reveal about how ALL waves behave? Your response must: (a) identify at least two wave properties or behaviours that are common to both sound waves and electromagnetic waves, supported by evidence from both mysteries; (b) explain how the wave equation v = fλ acts as a unifying tool that connects both explanations; (c) discuss one additional wave behaviour from this unit (such as reflection, refraction, or interference) and explain how it also applies to both sound and electromagnetic waves using a Kenyan example; and (d) answer the driving question directly and completely in 3–5 sentences.	explained the tunnel blackout, and the concept of frequency explained the pitch shift from the matatu. Second, both waves are affected by their physical environment in ways that depend critically on wavelength — the Doppler Effect depends on how the motion of the matatu compresses or stretches the sound wavelength, and the tunnel blackout depends on how the FM wavelength compares to the tunnel's dimensions. A third behaviour, reflection, applies equally to both: sound waves reflect off the concrete walls of buildings in Nairobi's CBD to create echoes, and radio waves reflect off the ionosphere above the Rift Valley, which is why some AM stations can be heard hundreds of kilometres from their transmitters. Taken together, the two mysteries directly answer the driving question: the matatu horn changes pitch because motion compresses and stretches sound wavelengths (the Doppler Effect), and the FM signal disappears in the Kabete tunnel because its short wavelength is absorbed and cannot diffract through the confined space — while the AM signal's enormous wavelength allows it to pass through. Both phenomena are governed by the same universal wave properties: frequency, wavelength, wave speed, and the behaviours of reflection, diffraction, and the Doppler Effect. Understanding these properties means you can explain and predict how any wave — sound, light, radio, water — will behave in any situation, from a highway in the Rift Valley to a radio tower in Turkana County.

Section 5: Applying Your Understanding — New Scenario in a Kenyan Context	
Demonstrate that your understanding goes beyond memorising the two original phenomena. A radio engineer working for a community FM station in Turkana County (broadcasting at 98.4 MHz) wants to ensure that her signal reaches villages on the far side of a rocky hill near Lodwar. She is also advising the Nairobi County government on why ambulance sirens seem louder and higher-pitched as they approach an accident scene on Thika Superhighway. Using what you have learned about wave properties in this unit: (a) advise the radio engineer on	For the Turkana FM engineer: the station broadcasts at 98.4 MHz, giving a wavelength of $\lambda = (3 \times 10^8) \div (98.4 \times 10^6) \approx 3.05$ m. This wavelength is extremely short compared to the dimensions of a rocky hill, which might be tens or hundreds of metres wide. Diffraction is only significant when a wave's wavelength is comparable to the size of the obstacle — since 3 m is far smaller than the hill, the FM signal will not diffract meaningfully over or around it. The signal will largely be blocked. A practical solution is to install a signal repeater or relay transmitter on top of the hill, which receives the signal on one side and re-broadcasts it to the villages on the other side. Alternatively, the engineer could apply for a frequency in the AM band (much longer wavelength) if high-quality audio fidelity is not the priority, since AM waves would diffract around the hill far more effectively. For the Nairobi County government: as an ambulance travels at speed along Thika Superhighway toward an accident scene, its siren emits sound waves that are compressed in front of it — raising the frequency and therefore the pitch heard by people ahead. Once it passes, the sound waves are stretched behind it, lowering the pitch. The speed of the ambulance determines how dramatic the shift is; a faster ambulance produces a more noticeable pitch change. Regarding limitations: the models used in this unit treated waves as behaving in ideal, uniform media, but real environments — like the Kabete tunnel with its irregular concrete surfaces, or the Rift Valley with its varied topography — create complex patterns of reflection, diffraction, and interference happening simultaneously. Our models did not fully account for these combined effects or for the role of temperature gradients in changing the speed of sound on a hot Nakuru afternoon. Further investigation using signal-strength sensors inside the tunnel, or computer wave-simulation software modelling the Turkana hill terrain, would give more precise and realistic predictions.

whether diffraction will help her FM signal travel over or around the hill, and suggest one practical solution she might consider — support your advice with wave property reasoning; (b) explain to the county government why the ambulance siren sounds change using the Doppler Effect; and (c) identify one limitation of your explanations — what did this unit's models NOT fully account for, and what further investigation might be needed?	

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Scientific Accuracy of Wave Properties	All wave properties (amplitude, wavelength, frequency, wave speed) are defined correctly with appropriate units. $v = f\lambda$ is applied accurately in at least two contexts, including correct numerical calculations for FM and AM wavelengths. Transverse and longitudinal waves are correctly distinguished with accurate examples from the phenomenon.	Most wave properties are defined correctly and $v = f\lambda$ is applied in at least one context. Minor errors in units or definitions do not obscure overall understanding. The distinction between transverse and longitudinal waves is present but may lack a strong link to the phenomenon.	Wave properties are partially defined or contain significant errors. $v = f\lambda$ is stated but not correctly applied to the phenomenon. Transverse and longitudinal waves may be confused or missing. Calculations, if attempted, contain errors that affect the explanation.
Explanation of the Doppler Effect (Matatu Horn Mystery)	The Doppler Effect is correctly named and fully explained: relative motion compresses wavelength in front of the source (raising frequency/pitch) and stretches it behind (lowering frequency/pitch). Explanation is clearly linked to the matatu scenario with correct reasoning about compression and wavelength change. An additional Kenyan example is provided and correctly explained.	The Doppler Effect is named and the basic idea of pitch change with motion is correctly described. The link between motion, wavelength change, and frequency is present but may not be fully developed. A second example may be present but weakly connected.	The pitch change is noted but not correctly attributed to wavelength compression or the Doppler Effect. Explanation may conflate loudness with pitch, or describe the driver changing the horn. Scientific vocabulary is largely absent or misused.
Explanation of Wavelength and Wave Behaviour in the Kabete Tunnel (FM vs AM Mystery)	FM and AM wavelengths are correctly calculated using $v = f\lambda$ and $c = 3 \times 10^8$ m/s. The role of wavelength relative to tunnel dimensions is clearly explained.	FM and AM wavelengths are compared correctly (even if not fully calculated). The basic idea that FM's short wavelength is blocked and AM's long	FM and AM are noted as different but the explanation relies on labels ('FM is weaker') rather than wave properties. Wavelength is not calculated or is

	Diffraction, absorption, and/or reflection are correctly applied to explain why FM is blocked and AM passes through. The explanation correctly identifies wavelength — not broadcast technology — as the key variable.	wavelength is not is present. At least one wave behaviour (diffraction, reflection, or absorption) is correctly applied. The explanation may not fully distinguish wavelength from technology type.	calculated incorrectly. Wave behaviours such as diffraction are mentioned but not correctly applied to the tunnel context.
Synthesis and Direct Answer to the Driving Question	The final explanation explicitly and directly answers the driving question in full. At least two wave properties are identified as common to both sound and electromagnetic waves, with evidence drawn from both mysteries. The wave equation is used as a unifying tool. The response demonstrates understanding that sound and radio waves obey the same fundamental principles despite their differences.	The driving question is addressed, and both mysteries are connected to shared wave properties. The unifying role of $v = f\lambda$ is acknowledged. The response may lack full integration — the two mysteries are explained separately rather than woven into one coherent argument, but the connection is still clear.	The driving question is partially answered or addressed only for one of the two mysteries. Wave properties are listed rather than used to build a unified argument. The connection between the two phenomena and shared wave behaviour is unclear or absent.
Application to Novel Kenyan Context and Identification of Limitations	The Turkana FM scenario is correctly analysed using wavelength and diffraction reasoning, with a practical and scientifically justified solution offered. The ambulance Doppler scenario is accurately explained. At least one genuine limitation of the unit's models is identified and a specific, feasible further investigation is proposed.	The Turkana FM scenario is addressed with mostly correct reasoning about wavelength and diffraction. A solution is suggested but may not be fully justified. The ambulance scenario is correctly explained. A limitation is identified but may be vague or not paired with a specific further investigation.	The novel scenarios are addressed with limited or incorrect use of wave property reasoning. Solutions offered are not grounded in wave properties. Limitations, if mentioned, are superficial (e.g., 'we need more time to study') and no meaningful further investigation is proposed.